

So You Want To Host An AWS Hackathon

Michael Geiser

BUSINESS VALUE OF HACKATHONS

Hackathons serve as catalysts for innovation, talent cultivation, and product development essential for successful client engagements in the Gen AI era.

Talent Development

Skill Enhancement

- Hands-on experience with cutting-edge technologies.
- Improvement in collaboration, communication, and project management skills.

Employee Engagement

- Excitement boosts morale and job satisfaction.
- Opportunities for employees to showcase their skills and creativity.

Career Path and Development

- Identification of high-potential employees.
- Cultivation of a pipeline of future leaders and innovators.

Enhancing Client Engagement

Prototype Development

- Quick iteration and validation of product ideas.
- Early-stage testing with real-world feedback.

Market-Ready Solutions Pipeline

- Transition from prototypes to fully-fledged products.
- Alignment with client needs and industry standards

Building Competitive Advantage

- Differentiation through unique, innovative offerings.
- Enhanced reputation as a forward-thinking, client-centric organization.

The First Hackathon - 2024

Bedrock Based Gen AI Hackathon

September 16 - September 30

\$26,627 in AWS costs offset by Service Credits

\$395 in 3P LLM usage

Additional ~\$7k in October because account termination was slow and labor intensive.

Many lessons learned, tool improvements, more efficient processes, and 3 AWS feature requests submitted.

1664 Registered Hackers

170 Teams

41 Countries



62 Submitted Projects

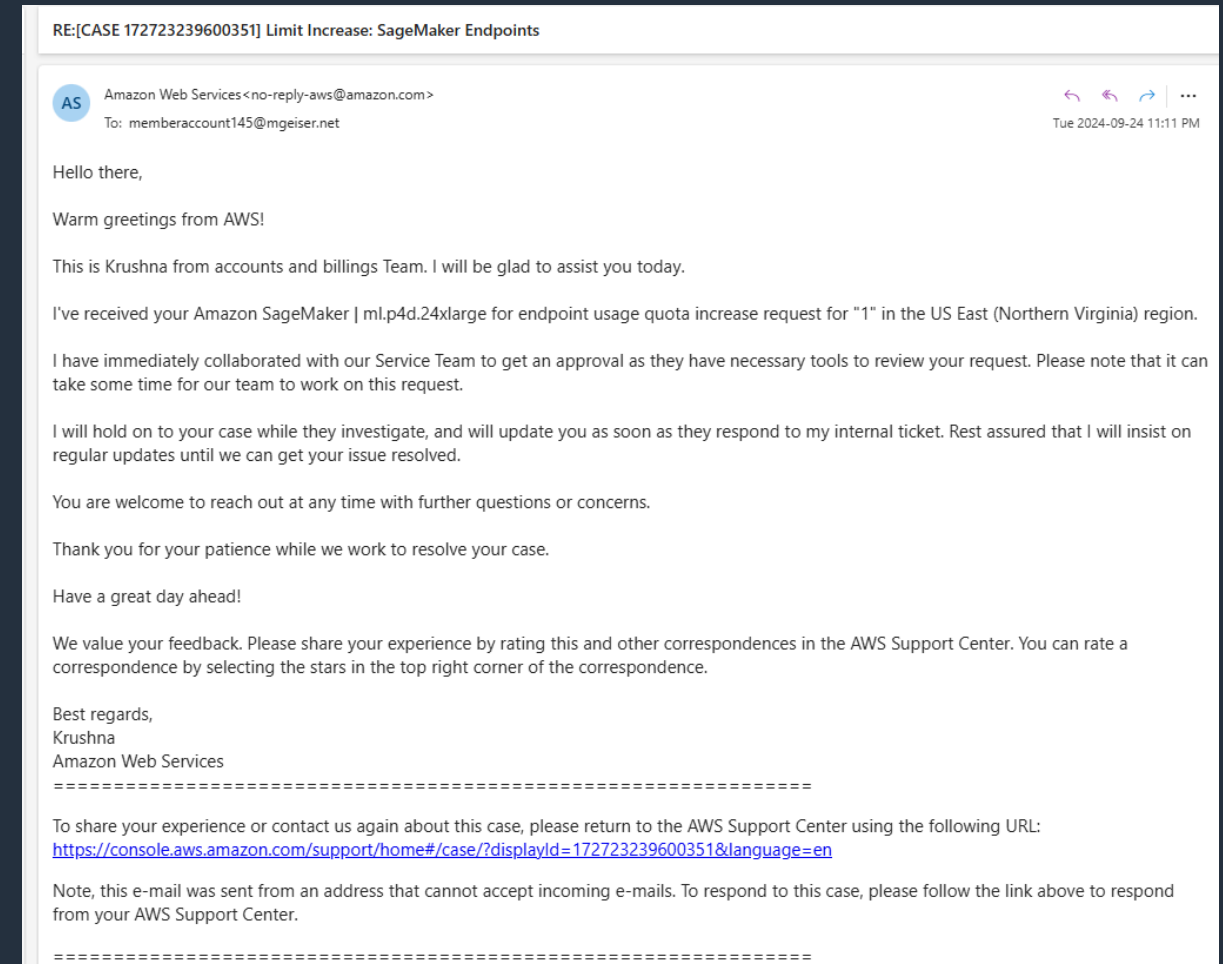
Expiration date	Credit name	Amount used	Amount remaining	Applicable products
01/31/2025	Q2 T&C Promo Credits- 500 v2	\$0.00	\$500.00	See complete list of services
01/31/2025	Q2 T&C Promo Credits- 500 v4	\$0.00	\$500.00	See complete list of services
01/31/2025	Q2 T&C Promo Credits- 500 v2	\$0.00	\$500.00	See complete list of services
01/31/2025	Q2 T&C Promo Credits- 500 v4	\$0.00	\$500.00	See complete list of services
01/31/2025	Q2 T&C Promo Credits- 500 v2	\$0.00	\$500.00	See complete list of services

Some of the misses:

First version of SCPs did not prevent all “undesired” actions.

Users could:

- Register domains
- Instantiate VERY large instances
- Sign up for support accounts
- Use AWS Marketplace products
- Purchase QuickSight subscriptions



The Second Hackathon - 2025

Agentic Gen AI Hackathon

July 7 – July 24

Rewrote vastly improved Account and User provisioning tools with Amazon Q

- Only 4 hours (including full testing)

Refined SCPs and Permissions Boundaries

Leveraged Control Tower features better

New AWS features reduced manual tasks

Twice the participants; about half the spend due to better controls

3261 Registered Hackers

404 Teams

45 Countries



132 Submitted Projects



GUARDRAILS ENSURE A SECURE HACKATHON

Organizational Controls:

- Hackathon accounts isolated in dedicated **AWS Organization OUs**.
- SCPs and IAM Permission Boundaries enforce limitations on entitlements.
- All accounts auto-tagged and monitored as ephemeral/non-production.

Identity & Access Management:

- No IAM Users allowed; **IAM Identity Center** role-based permissions only.
- Strict **Permission Boundary** limits enforced on all roles.
- SCPs deny privilege escalation, access to sensitive AWS services (IAM, Organizations, Account), and disallowed regions.

Network Controls:

- **Internet egress blocked** via SCPs.
- Internet access only allowed by moving accounts to OU with appropriate guardrails.
- Private VPC architecture encouraged, with optional VPC endpoints for GenAI APIs.

Data Protection:

- No access to customer or production data.
- Data uploads restricted to defined Amazon S3 buckets with lifecycle rules to delete content post-event.
- Amazon Bedrock used **Amazon models** and third-party models with special permission.
- All products/IP produced reverts to Capgemini.

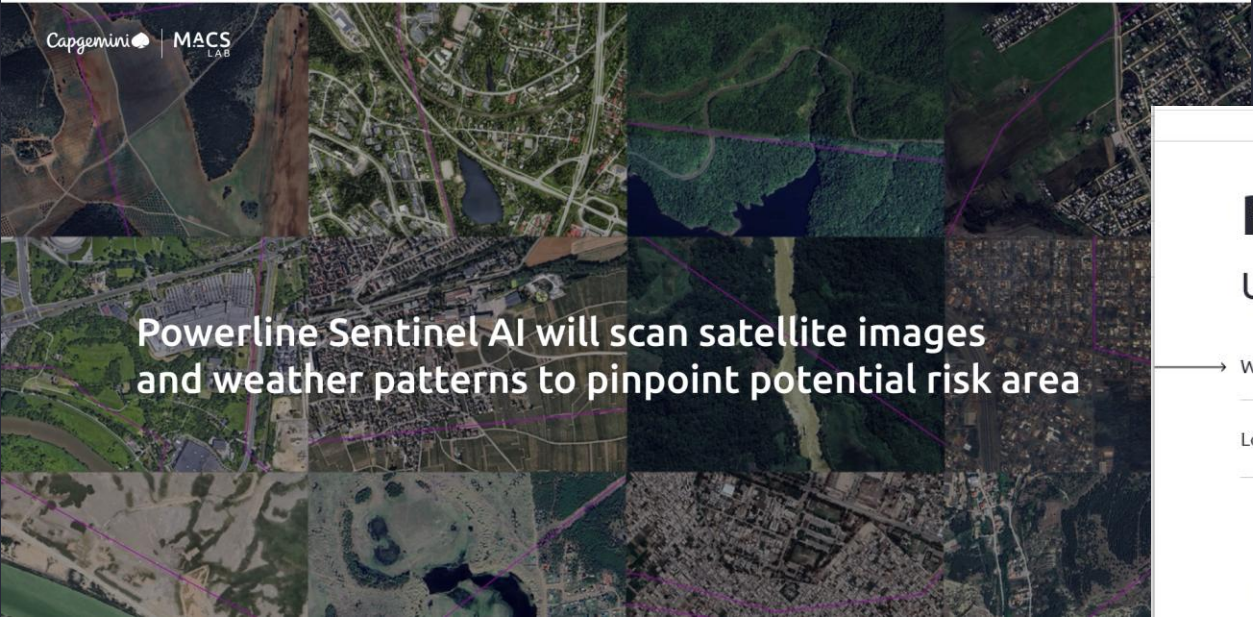
Let me say thank you while
finishing up my final point...

Experience, new skills, new features, and
feedback loops are critical to getting it right

- Separate from Corporate Dev, Test, and Production AWS accounts.
- Security Policies created and reviewed by Certified Security Architect and AWS SAs.
- All accounts hardened with Preventative SCPs and Permission Boundaries.
- Additional active Detective security processes in place with non-compliance action plans.
- Only whitelisted GenAI models and tools enabled.
- Audit logging and cost controls in place.
- No persistent cloud resources post-hackathon.

Appendix Material

2024 HACKATHON WINNER FEATURED AT AWS RE:INVENT 2024



Report for U.S. — California — Humboldt County

→ Wednesday, 18 June 2025

Location [40.483517,-123.781392]

Power Line HV – High Voltage

HIGH RISK

11.78 °C

Avg. temperature

71.78 RH2M

Avg. humidity

0.68

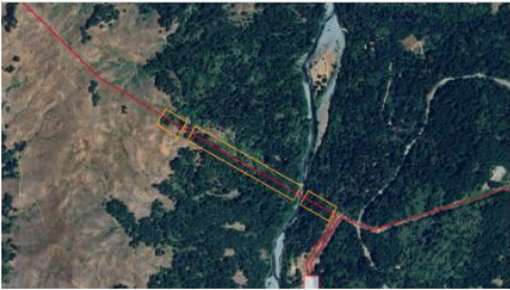
Soil moisture

35

Estimated growth rate

Risk assessment

The vegetation growth risk in this area is high. Despite an average growth rate, visible neglect in maintaining the plant state in detected areas increases the likelihood of uncontrolled vegetation spread. Immediate intervention and consistent maintenance are necessary to prevent further deterioration and encroachment.



FileEditSelectionViewGoRun...<=>BH-HackathonUtil

EXPLORER

BH-HACKATHONUTIL

archive

account_mgmt.md

account_mgmt.py

account_teamid_map.csv

accounts_to_close.csv

accounts_to_close.md

accounts_to_close.py

active_accounts_list.csv

available_accounts.csv

bh-hackers.csv

bh-hackers1.csv

Capgemini - Baker Hughes AWS Hackathon 2025 Instructions.docx

Capgemini - Baker Hughes AWS Hackathon 2025 Instructions.pdf

clean_nonteam_users.md

clean_nonteam_users.py

configuration.yaml

create_account_log_20250912_110618.csv

create_account_log_20250912_122446.csv

create_account_log_20250917_153429.csv

create_member_accounts.md

create_member_accounts.py

deprovision_users.md

deprovision_users.py

hackathon admin.7z

hackers-0.csv

run_log_20250912_125013.csv

run_log_20250917_162830.csv

OUTLINE

TIMELINE

POWER PAGES ACTIONS

APPLICATION BUILDER

create_member_accounts.py

create_member_accounts.py

validate_config

1import yaml

2import boto3

3import csv

4from datetime import datetime

5

6def load_config(config_file='configuration.yaml'):

7 with open(config_file, 'r') as file:

8 return yaml.safe_load(file)

9

10def validate_config(config):

11 """Validate configuration parameters before execution"""

12 required_params = []

13 'aws_access_key_id',

14 'aws_secret_access_key',

15 'sso_region',

16 'new_account_start_value',

17 'new_account_end_value',

18 'new_account_role_name',

19 'new_account_ou_target'

20]

21

22 # Check required parameters exist and are not empty

23 for param in required_params:

24 if param not in config or not config[param]:

25 print(f"ERROR: Missing or empty required parameter: {param}")

26 exit(1)

27

28 # Validate numeric ranges

29 start_val = config['new_account_start_value']

30 end_val = config['new_account_end_value']

31

32 if not isinstance(start_val, int) or not isinstance(end_val, int):

33 print("ERROR: new_account_start_value and new_account_end_val")

34 exit(1)

35

36 if start_val > end_val:

CHAT

+

🔍

⚙️

...

🗨️

×

Build with agent mode

AI responses may be inaccurate.

Generate Agent Instructions to onboard AI onto your codebase.

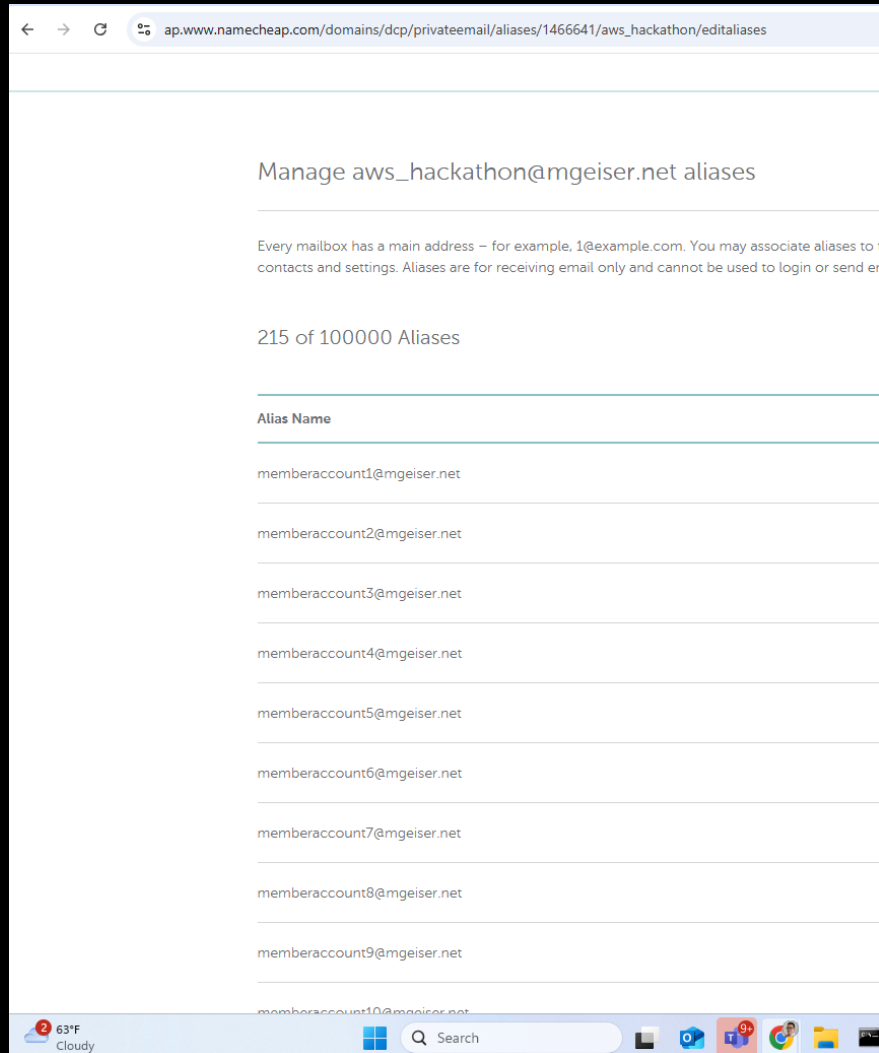
create_member_accounts.py +

Add context (#), extensions (@), α

Agent ▾ Auto ▾ 🔊 ▶ ▾

Ln 14, Col 34 Spaces: 4 UTF-8 LF {} Python 3.12.1

Aliases in accounts



Subaddressing accounts

You can also get additional unique email addresses (for example: for new AWS accounts) is to use subaddressing (defined in RFC-5233) this is also known as “plus addressing”, tagged addressing or mail extensions). This means that the email address user@example.com can use:

- user+management@example.com - for your management account
- user+logs@example.com - for your log archive account
- user+audit@example.com - for your audit account
- user+sandbox@example.com - for your sandbox account

All addresses are unique, valid, and would be delivered to the mailbox associated with user@example.com.

This works in Exchange:

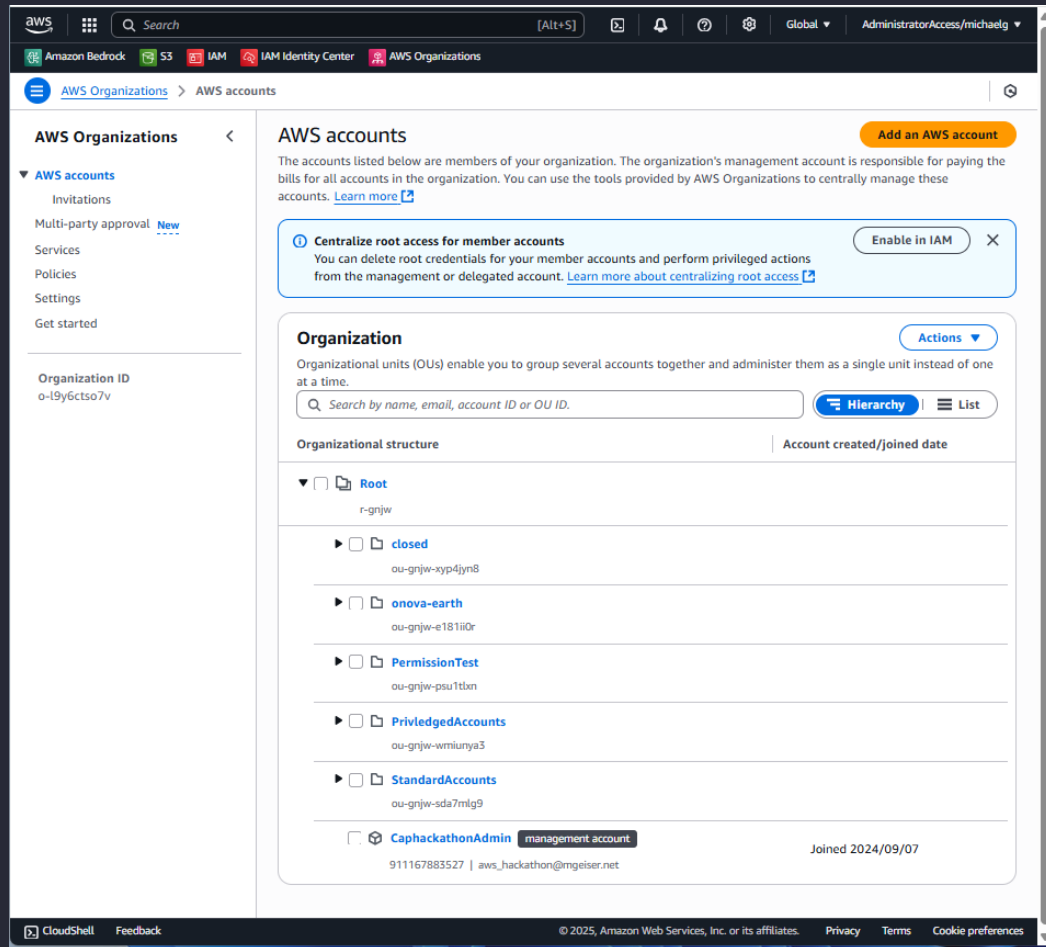


In Gmail:

- michael.j.geiser@gmail.com
- michaeljgeiser@gmail.com
- m.i.c.h.a.e.l.j.g.e.i.s.e.r@gmail.com

are all deliver mail to the same account; dashes and underscores do differentiate

ACCOUNT ORGANIZATION OVERVIEW



The AWS Organization OUs:

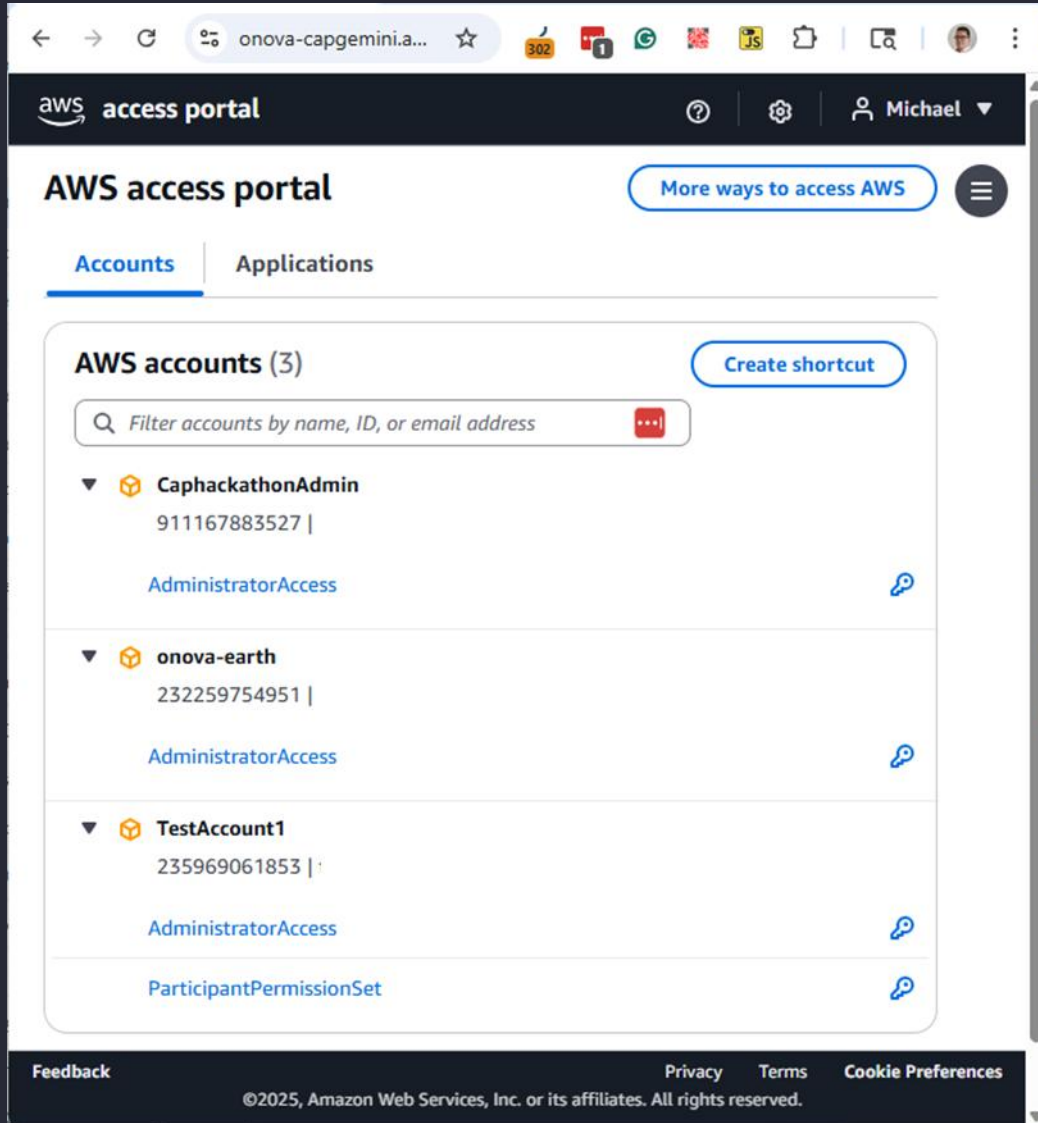
StandardAccounts: all Hackathon Member Accounts are placed here. A restrictive SCP and Permission Boundary Policy are applied to this OU.

PrivledgedAccounts: Account needing permissions beyond a Standard Account are moved to this OU with a modified SCP. 2 Member Accounts elevated in 2025.

PermissionTest: Custom SCPs are tested in this OU before deployed to other accounts.

closed: Accounts that closed are moved to this OU. The accounts are inaccessible because they are closed and all permissions via policy are essential “Deny *”

USER LOGIN ACCESS TO ALL ACCOUNTS VIA IAM IDENTITY CENTER



During the hackathon, the admins could add or subtract member accounts using low effort & logged functions.

Admin users can also have the **participant** permissions. Additional roles are available in the AWS Access Portal.

Hackathon users will only have **participant** permissions.

All participants are required to use MFA.

SERVICE CONTROL POLICY (SCP)

A Service Control Policy (SCP) enforces regional restrictions, limits high-risk actions, and restricts unapproved services and configurations to maintain security, compliance, and cost control across AWS accounts. The SCP for the Hackathon enforces these restraints

Category	Controls	Notes
Region Restriction	Denies all non-core actions outside us-east-1	IAM, Org, STS, Route53, etc. still allowed globally
Model & AI	Blocks Bedrock model provisioning/customization	Prevents GenAI spend and misuse
Analytics	Denies all Amazon QuickSight access	Helps avoid ungoverned BI sprawl
Marketplace	Denies acceptance of marketplace agreements	Prevents surprise costs and software liability
Domain Registration Services	Blocks users from registering domains	This is not needed and adds unnecessary costs
EC2 Usage	Allows only a limited set of instance types	Limits cost and ensures approved sizing
VPC Configuration	Denies creation/modification of VPCs and subnets	Network changes must be managed centrally
Root Account Security	Blocks sensitive IAM actions on root user	Enforces least privilege for high-risk identity
Container Services	Blocks usage of EKS and ECS	Prevents uncontrolled container deployments
Internet Access	Denies creation of IGWs, NATs, and elastic Ips	Restricts open outbound connectivity
SageMaker	Blocks public Mechanical Turk labeling jobs	Prevents data exposure via public crowdsourcing
S3 Public Access	Blocks bucket and object ACLs/policies that enable public access	Enforces private S3 usage policies
S3 Bucket Policies	Denies bucket policy changes without a defined principal and removal of bucket encryption.	Prevents risky anonymous or broad access
Search Service	Blocks all OpenSearch Service actions (off for this hackathon)	Restricts unapproved search deployments

PERMISSION BOUNDARIES

A Permissions Boundary policy limits privileged IAM actions while allowing controlled role creation. It ensures participants can experiment safely without escalating privileges or bypassing key restrictions.

Category	Controls	Notes
IAM Restrictions	Denies all iam:*, organizations:*, and account:* actions	Blocks privilege escalation and org-level changes
Controlled Role Creation	Allows iam:CreateRole only if HackathonBoundaryPolicy is attached	Ensures new roles stay within boundary limits
Boundary Enforcement	Denies removal of role permission boundaries	Prevents elevation by boundary deletion
Policy Protection	Denies tampering with the boundary policy (detach, delete, versioning)	Locks down the boundary itself
Sensitive Service Denial	Denies OpenSearch and public SageMaker labeling jobs (Mechanical Turk)	Blocks unapproved data exposure or search service usage

Outlook

Search

HomeViewHelp

New

DeleteArchive

Report

Reply

Reply all

Forward

Meeting

Move

Read / Unread

Flag

Pin

Snooze

Print

Discover groups

Undo

Capgemini/Sogeti - AWS Dat

Now, https://chime.aws/4514

New

Delete

Report

Respond

Move

Tags

Print

Find

Undo

Favorites

michael.geiser@capgemini.c...

mgeiser.workshop@gmail.com

aws_account_hp@mgeiser.net

aws_hackathon@mgeiser.net

Inbox

32

action needed

Closed accounts

107

IAM IDC

Invoices

processed

2

Rajesh

Junk Email

Drafts

Sent Items

Deleted Items

19

Archive

Outbox

Add account

FocusedOther

By Date

Other Emails (3)

Amazon Web Services

Amazon Web Services

AWS Account Permanent Closure Confirmation

Tue 04-29

To whom it may concern, This e-mail confirms that the Amazon Web Services account ...

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Amazon Web Services

AWS Account Permanent Closure Confirmation

Tue 04-29

To whom it may concern, This e-mail confirms that the Amazon Web Services account ...

AWS Account Permanent Closure Confirmation

ASAmazon Web Services<no-reply@amazonaws.com>

To: memberaccount35@mgeiser.net

Tue 2025-04-29 4:25 PM

aws

To whom it may concern,

This e-mail confirms that the Amazon Web Services account associated with account ID 911167919290 is permanently closed and cannot be reopened. Any content remaining in this account is inaccessible and will be erased.

Sincerely,
Amazon Web Services

Amazon Web Services, Inc. is a subsidiary of Amazon.com, Inc. Amazon.com is a registered trademark of Amazon.com. This message was produced and distributed by Amazon Web Services, Inc. or its affiliates, 410 Terry Ave. North, Seattle, WA 98109.

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Reply

Forward



us-east-1.console.aws.amazon.com/organizations/v2/home/accounts

Search [Alt+S]

Global AdministratorAccess/michaelg

AWS Organizations

AWS accounts

AWS Organizations

AWS accounts

Invitations

Multi-party approval

Services

Policies

Settings

Get started

Organization ID

o-l9y6ctso7v

Organizational structure

Account created/joined date

Root

r-grjw

closed

ou-grjw-xyp4jyn8

onova-earth

ou-grjw-e181i0r

PermissionTest

ou-grjw-psu1tdm

PrivledgedAccounts

ou-grjw-wmilunya3

StandardAccounts

ou-grjw-sda7mlg9

ParticipantAcct134

173587200717 | hackathonaccount134@imgeliser.net

Created 2025/07/04

ParticipantAcct14

169164939824 | hackathonaccount14@imgeliser.net

Created 2025/07/04

ParticipantAcct210

486261218466 | hackathonaccount210@imgeliser.net

Created 2025/07/07

ParticipantAcct212

137918335733 | hackathonaccount212@imgeliser.net

Created 2025/07/07

ParticipantAcct223

120671645385 | hackathonaccount223@imgeliser.net

Created 2025/07/07

ParticipantAcct88

700526301103 | hackathonaccount88@imgeliser.net

Created 2025/07/04

CaphackathonAdmin

911167883527 | aws_hackathon@imgeliser.net

Joined 2024/09/07

us-east-1.console.aws.amazon.com/organizations/v2/home/accounts

Search [Alt+S]

Global AdministratorAccess/michaelg

AWS Organizations

AWS accounts

AWS Organizations

AWS accounts

Invitations

Multi-party approval

Services

Policies

Settings

Get started

Organization ID

o-l9y6ctso7v

AWS accounts

Add an AWS account

The accounts listed below are members of your organization. The organization's management account is responsible for paying the bills for all accounts in the organization. You can use the tools provided by AWS Organizations to centrally manage these accounts. [Learn more](#)

Centralize root access for member accounts

You can delete root credentials for your member accounts and perform privileged actions from the management or delegated account. [Learn more about centralizing root access](#)

Enable in IAM

Organization

Organizational units (OUs) enable you to group several accounts together and administer them as a single unit instead of one at a time.

Search by name, email, account ID or OU ID.

Hierarchy

List

Organizational structure

Account created/joined date

Root

r-grjw

closed

ou-grjw-xyp4jyn8

ParticipantAccount105

205930637957 | memberaccount105@imgeliser.net

Created 2024/09/18

ParticipantAccount125

209479307705 | memberaccount125@imgeliser.net

Created 2024/09/18

ParticipantAccount143

148761672708 | memberaccount143@imgeliser.net

Created 2024/09/18

ParticipantAccount164

216989101395 | memberaccount164@imgeliser.net

Created 2024/09/25

ParticipantAccount167

195275664488 | memberaccount167@imgeliser.net

Created 2024/09/26

ParticipantAccount170

122610522724 | memberaccount170@imgeliser.net

Created 2024/09/26

ParticipantAccount177

135808931903 | memberaccount177@imgeliser.net

Created 2024/09/26

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aws

Search

[Alt+S]

Amazon Bedrock

S3

IAM

IAM Identity Center

AWS Organizations

Lambda

Billing and Cost Manage...

IAM Identity Center

Users

IAM Identity Center

<

Managing instance

Onova-Capgemini-Hackathon

Dashboard

Users

Groups

Settings

Multi-account permissions

AWS accounts

Permission sets

Application assignments

Applications

Related consoles

CloudTrail

Recommended

AWS Organizations

IAM

Users (1246)

Users listed here can sign in to the AWS access portal to access AWS accounts and assigned cloud applications.[Learn more](#)

Username

Find users

Delete users

Add user

<

1

2

3

4

5

...

>

Settings

☐	Username	Display name	Status	MFA devices	Created by
☐	michaelg	Michael Geiser	Enabled	1 device	Manual
☐	molka.el-jazi@capgemini.com	molka.el-jazi@capgemini.com	Enabled	1 device	Manual
☐	senthil.tr@capgemini.com	senthil.tr@capgemini.com	Enabled	1 device	Manual
☐	candice.legeay@capgemini.com	candice.legeay@capgemini.com	Enabled	1 device	Manual
☐	smita.a.kumar@capgemini.com	smita.a.kumar@capgemini.com	Enabled	1 device	Manual
☐	mukut.roy@capgemini.com	mukut.roy@capgemini.com	Enabled	1 device	Manual
☐	kaushal.f.kumar@capgemini.com	kaushal.f.kumar@capgemini.com	Enabled	1 device	Manual
☐	swaminathan.govindaramanuja...	swaminathan.govindaramanujam-ra...	Enabled	1 device	Manual
☐	abhishek-kumar.gupta@capge...	abhishek-kumar.gupta@capgemini....	Enabled	1 device	Manual
☐	harsha-sai.vaddem@capgemini....	harsha-sai.vaddem@capgemini.com	Enabled	1 device	Manual
☐	fulin.huang@capgemini.com	fulin.huang@capgemini.com	Enabled	1 device	Manual
☐	sejal.a.shah@capgemini.com	sejal.a.shah@capgemini.com	Enabled	1 device	Manual
☐	akhilesh-anil.shingare@capgem...	akhilesh-anil.shingare@capgemini.c...	Enabled	None	Manual
☐	mo.uddin@capgemini.com	mo.uddin@capgemini.com	Enabled	1 device	Manual
☐	praneeth.a.kumar@capgemini.c...	praneeth.a.kumar@capgemini.com	Enabled	None	Manual
☐	phanendra.samavedula@capge...	phanendra.samavedula@capgemini....	Enabled	1 device	Manual
☐	karthiga2.r@capgemini.com	karthiga2.r@capgemini.com	Enabled	1 device	Manual
☐	kumar.f.abhishek@capgemini.com	kumar.f.abhishek@capgemini.com	Enabled	1 device	Manual
☐	saumya.a.ranjan@capgemini.com	saumya.a.ranjan@capgemini.com	Enabled	1 device	Manual
☐	praveen.perumal-pillai@capge...	praveen.perumal-pillai@capgemini....	Enabled	None	Manual

https://us-east-1.console.aws.amazon.com/console/home?region=us-east-1

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